

General Segmentation Pipeline Development for Microscopy Images with Complex Cell Morphologies

Appendix

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1. APPENDIX 1 – BASE IMAGE SEGMENTATION RESULTS

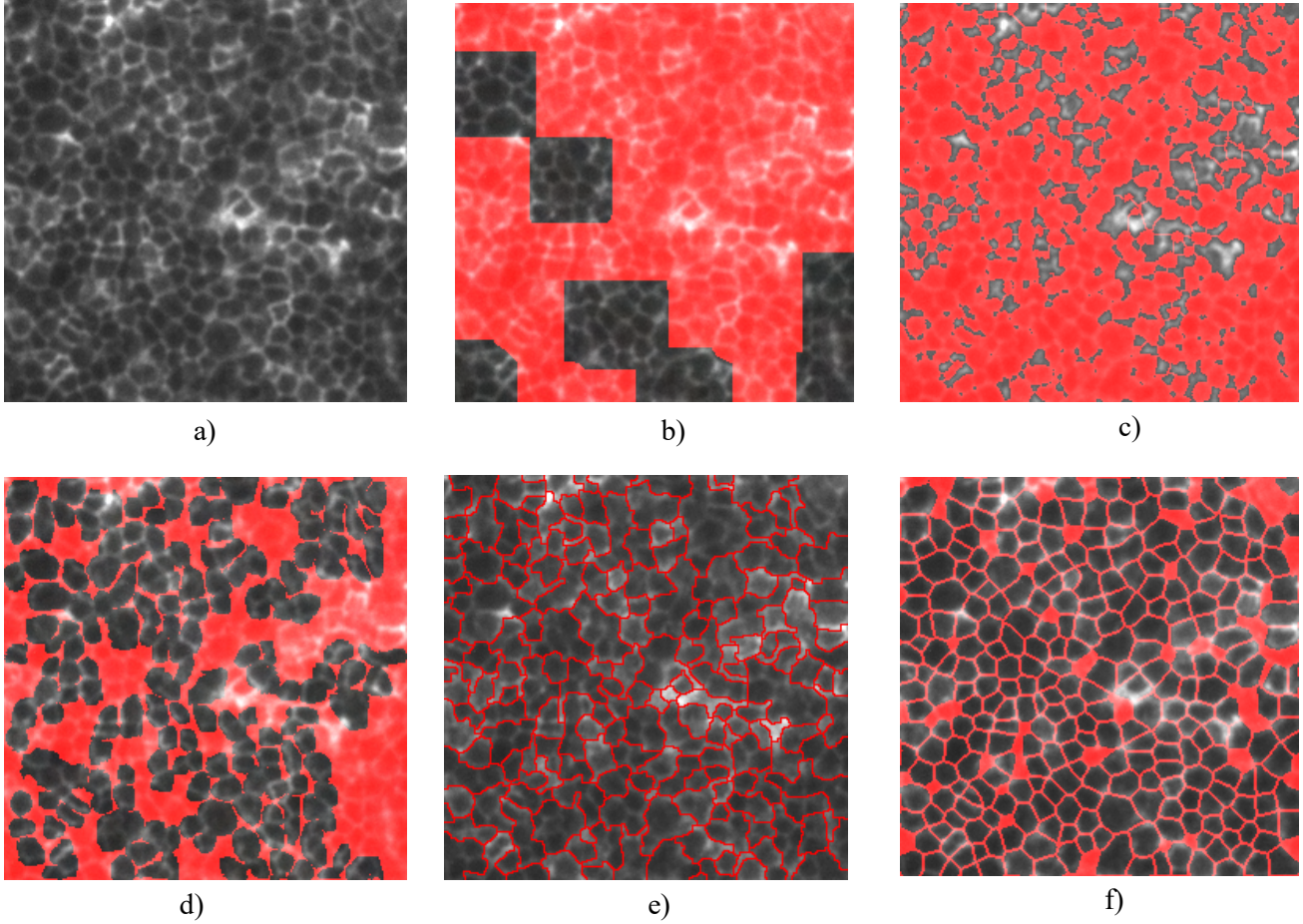
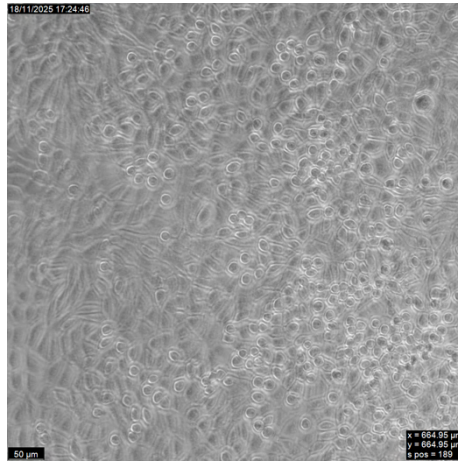
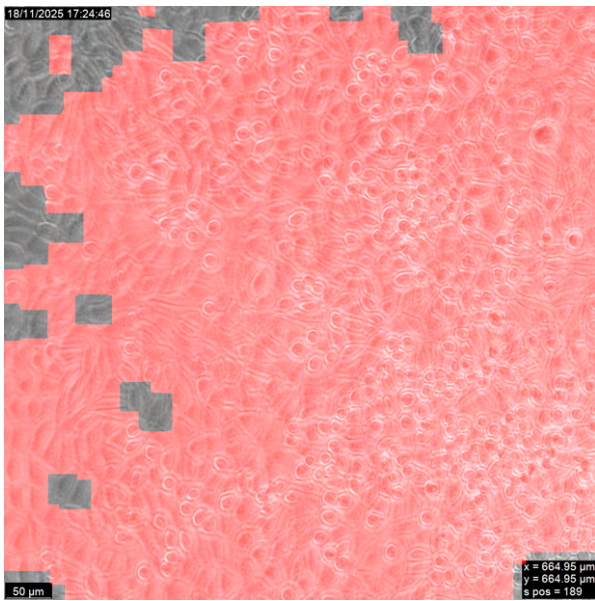


Figure 1. Base Image segmentation results. a) Original image, b) Intensity Thresholding, c) ImageJ Sobel + Watershed, d) Mask R-CNN, e) Developed Sobel+Watershed, f) Cellpose-SAM

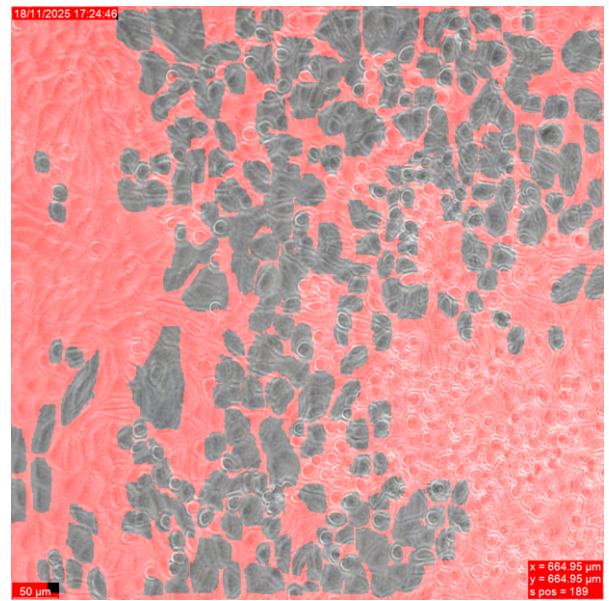
2. APPENDIX 2 – F1 SEGMENTATION RESULTS



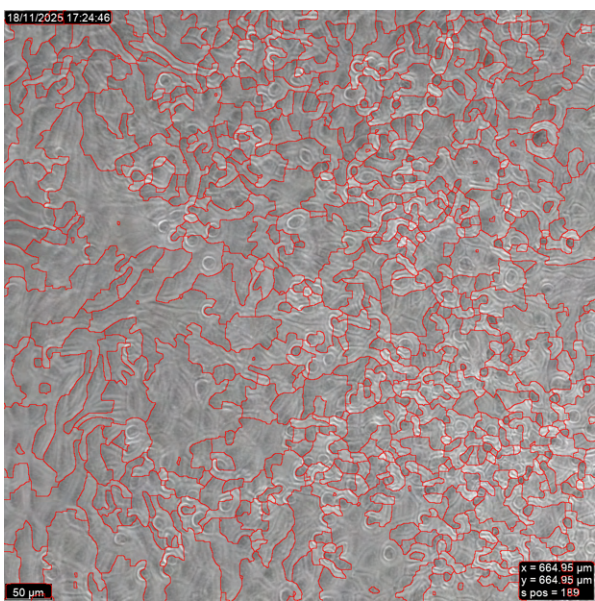
a)



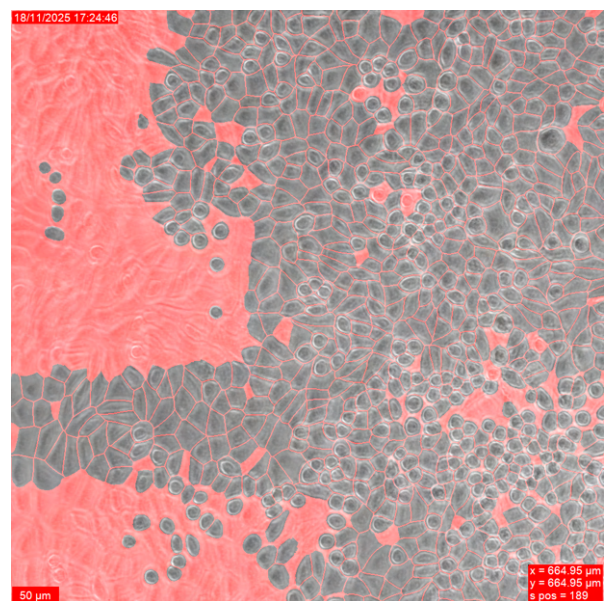
b)



c)



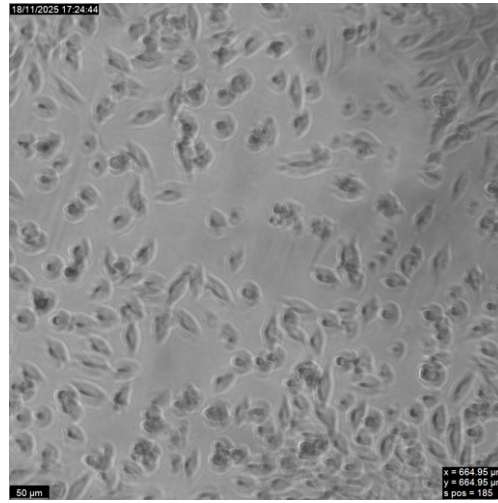
d)



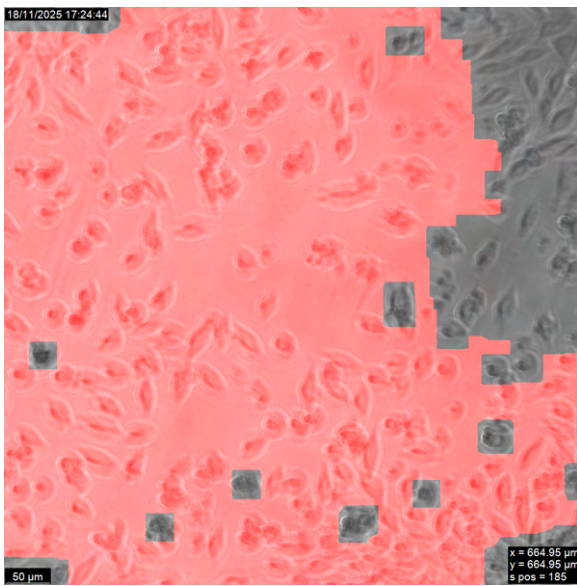
e)

Figure 2. F1 results. a) Original image, b) Intensity Thresholding, c) ImageJ Sobel + Watershed, d) Mask R-CNN, e) Developed Sobel+Watershed, f) Cellpose-SAM

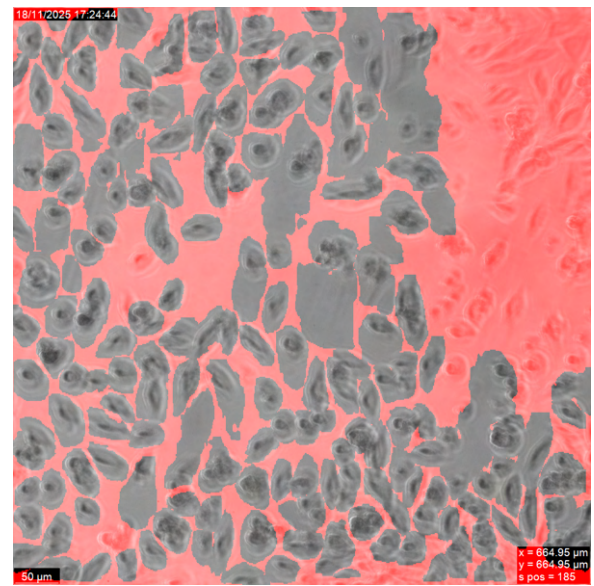
3. APPENDIX 3 – F2 SEGMENTATION RESULTS



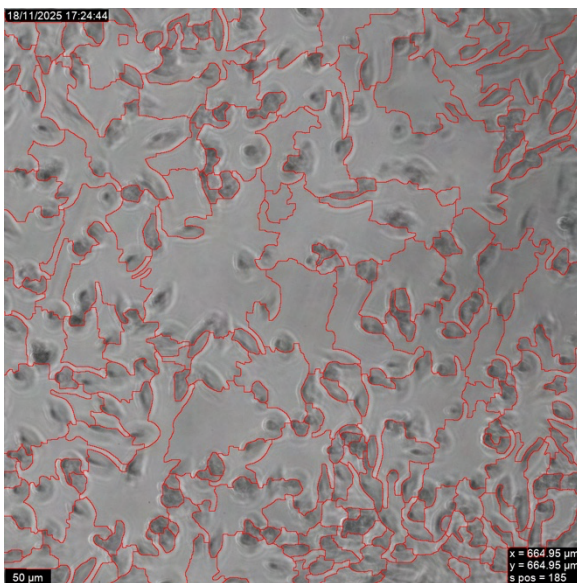
a)



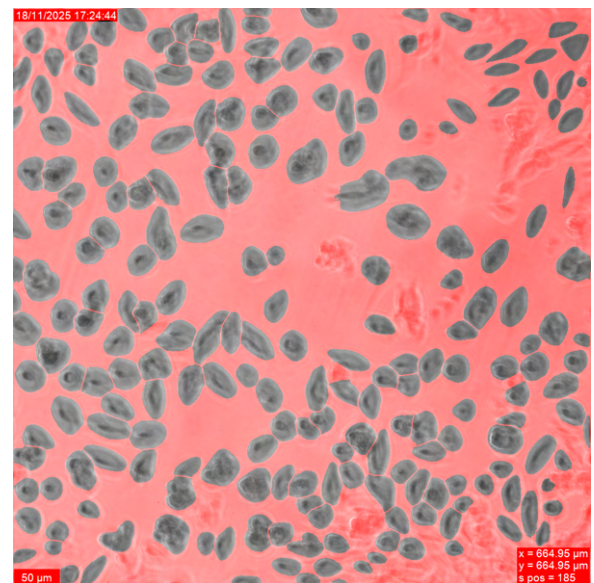
b)



c)



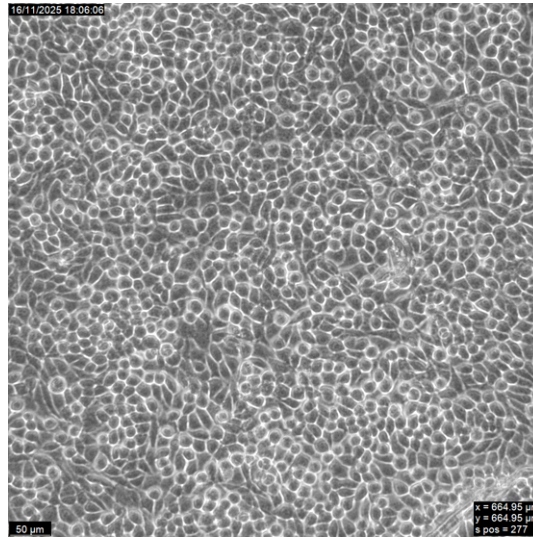
d)



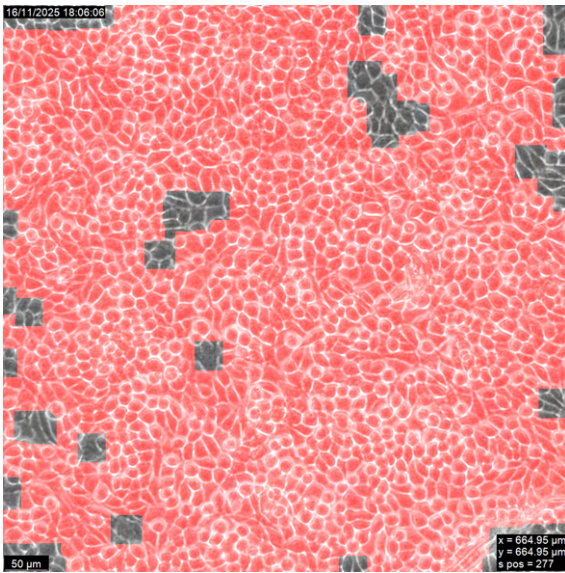
e)

Figure 3. F2 results. a) Original image, b) Intensity Thresholding, c) ImageJ Sobel + Watershed, d) Mask R-CNN, e) Developed Sobel+Watershed, f) Cellpose-SAM

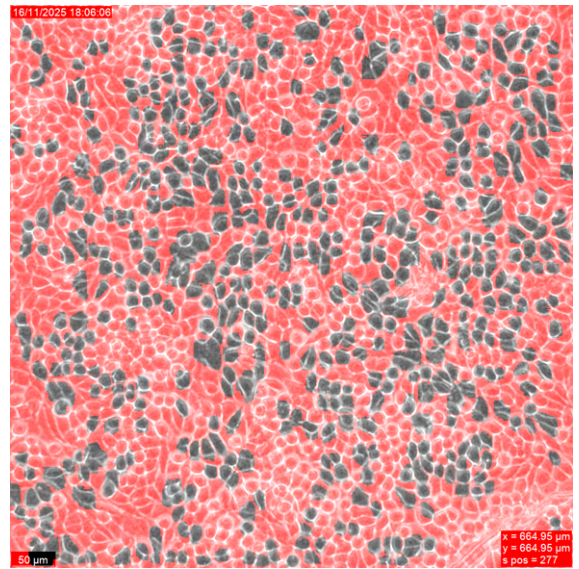
4. APPENDIX 4 – F3 SEGMENTATION RESULTS



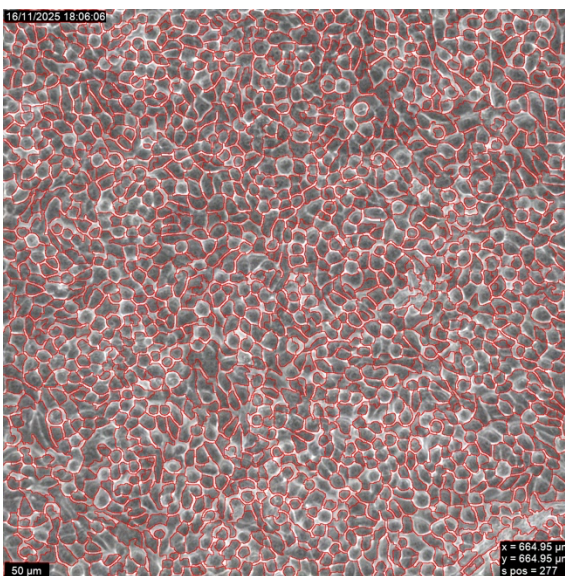
a)



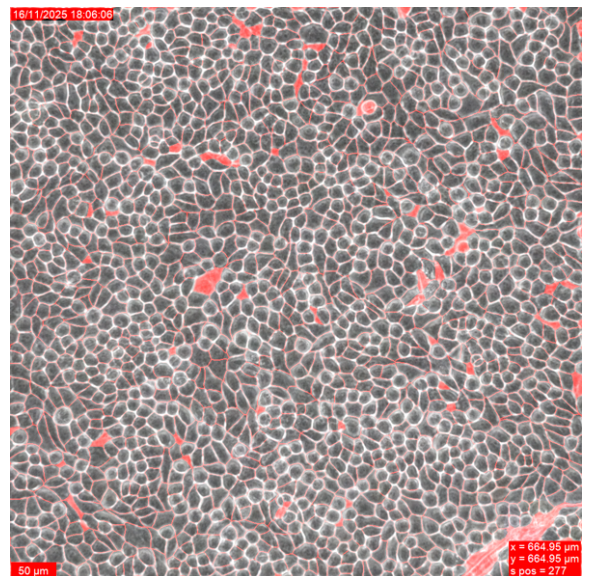
b)



c)



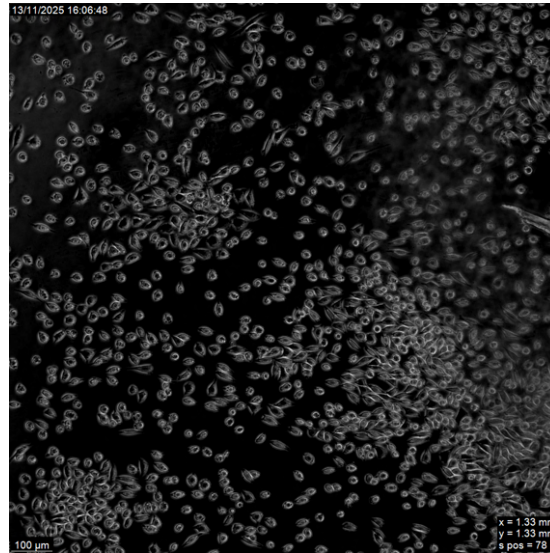
d)



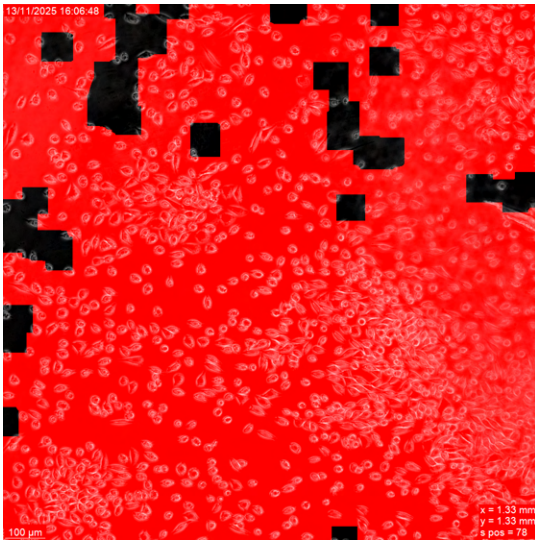
e)

Figure 4. F3 results. a) Original image, b) Intensity Thresholding, c) ImageJ Sobel + Watershed, d) Mask R-CNN, e) Developed Sobel+Watershed, f) Cellpose-SAM

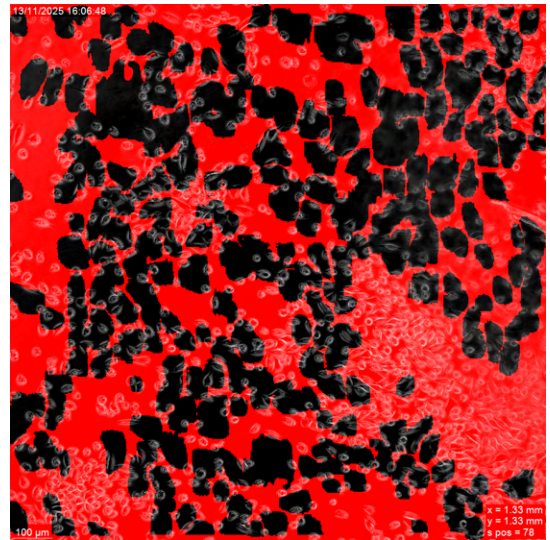
5. APPENDIX 5 – F4 SEGMENTATION RESULTS



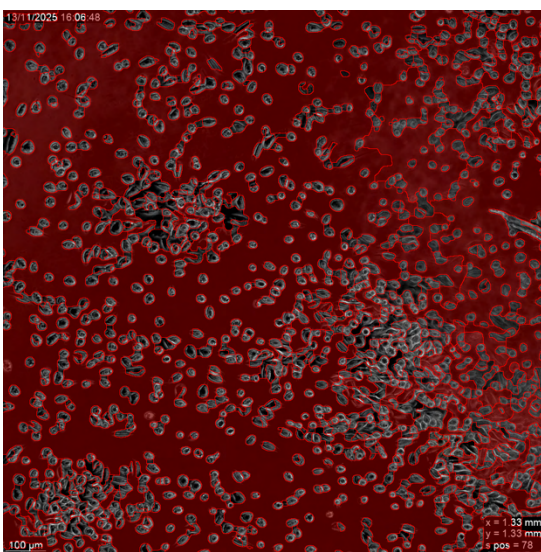
a)



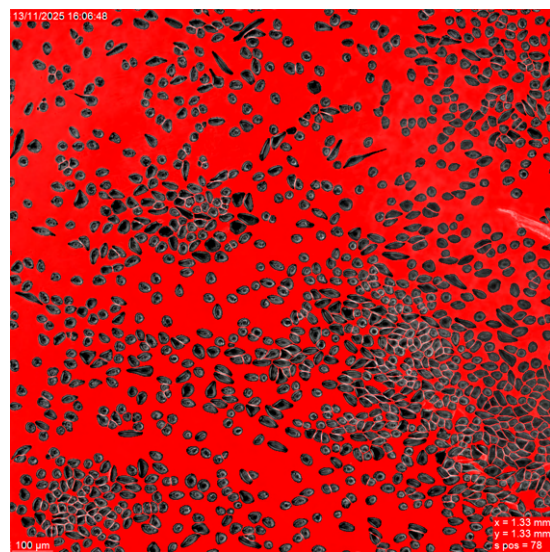
b)



c)



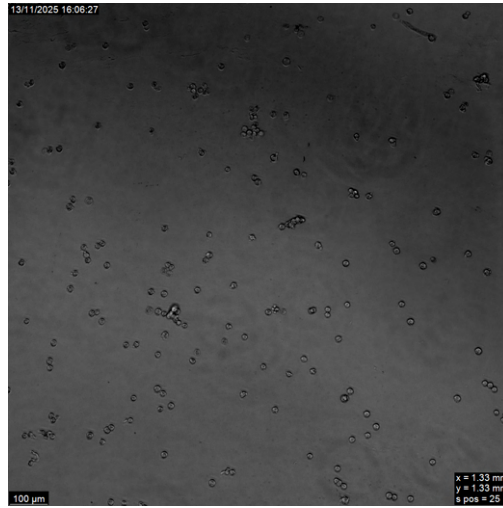
d)



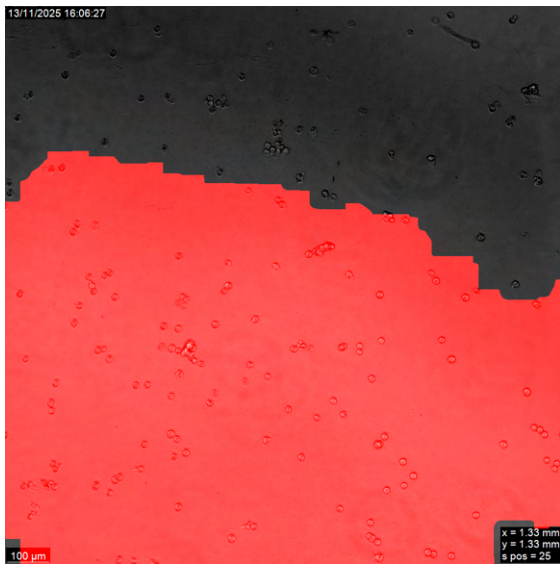
e)

Figure 5. F4 results. a) Original image, b) Intensity Thresholding, c) ImageJ Sobel + Watershed, d) Mask R-CNN, e) Developed Sobel+Watershed, f) Cellpose-SAM

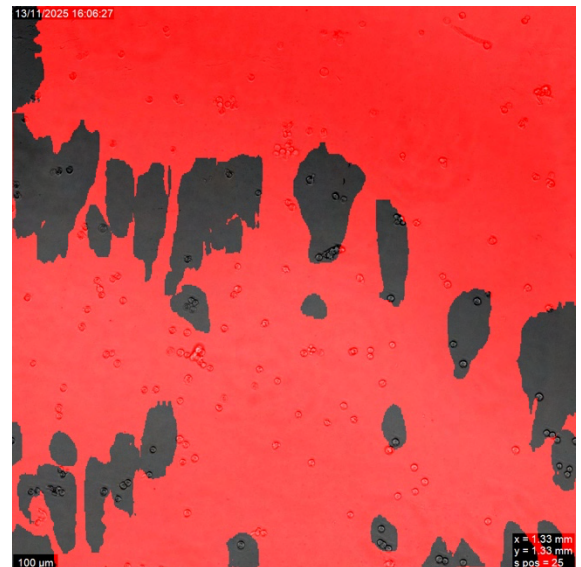
6. APPENDIX 6 – F5 SEGMENTATION RESULTS



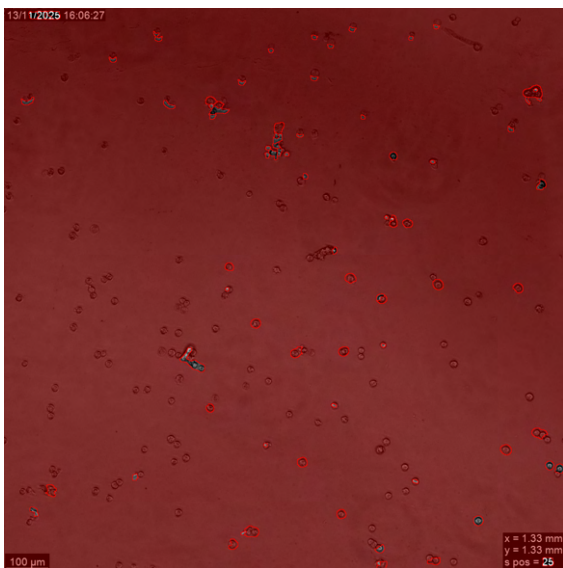
a)



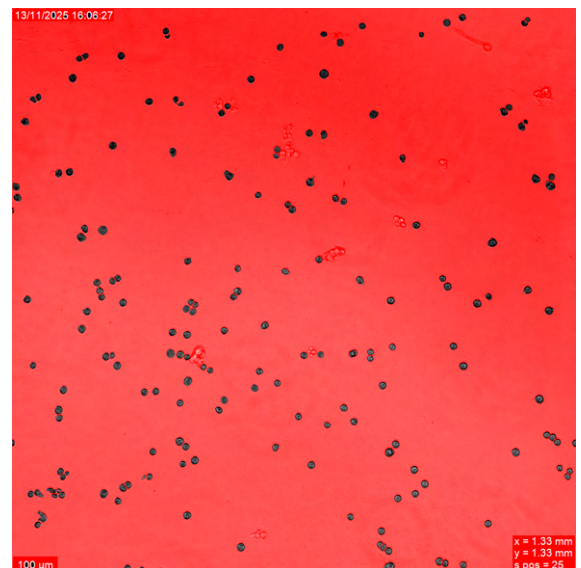
b)



c)



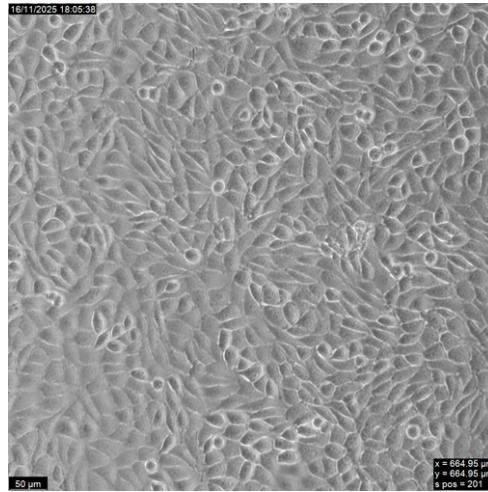
d)



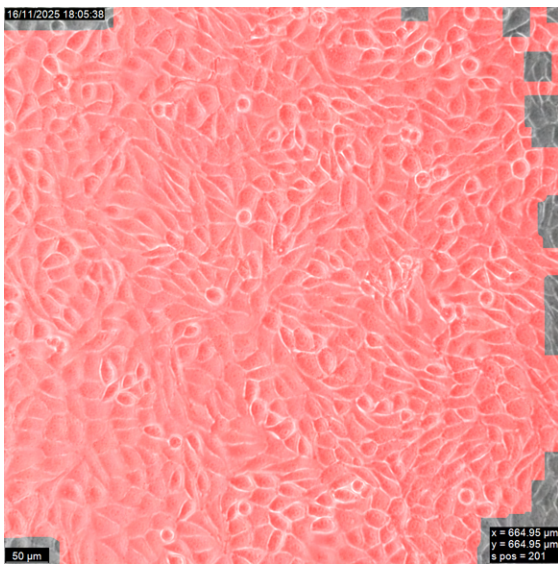
e)

Figure 6. F5 results. a) Original image, b) Intensity Thresholding, c) ImageJ Sobel + Watershed, d) Mask R-CNN, e) Developed Sobel+Watershed, f) Cellpose-SAM

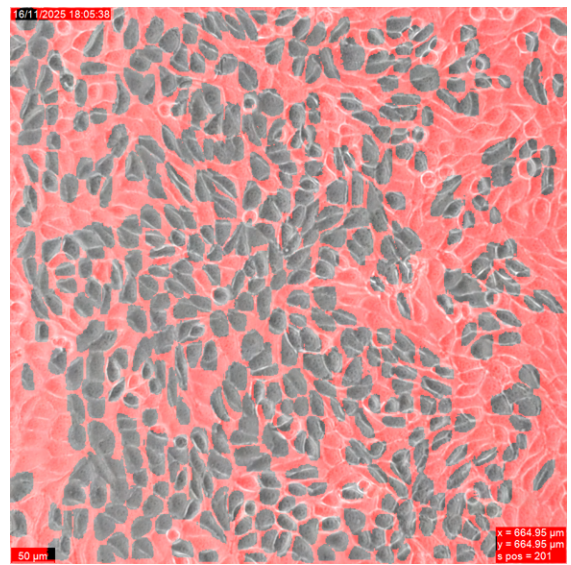
7. APPENDIX 7 – F6 SEGMENTATION RESULTS



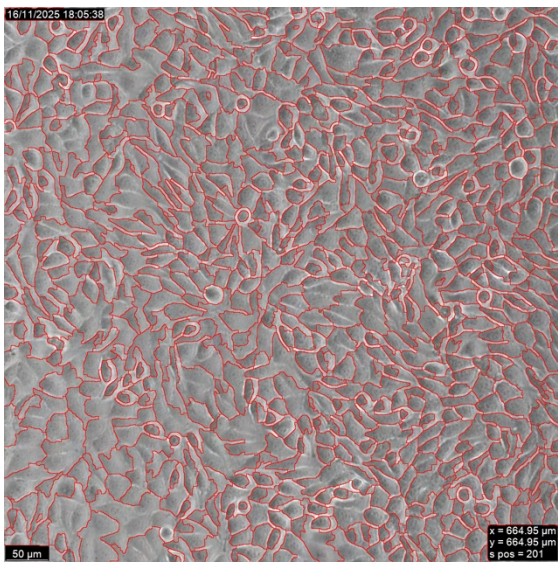
a)



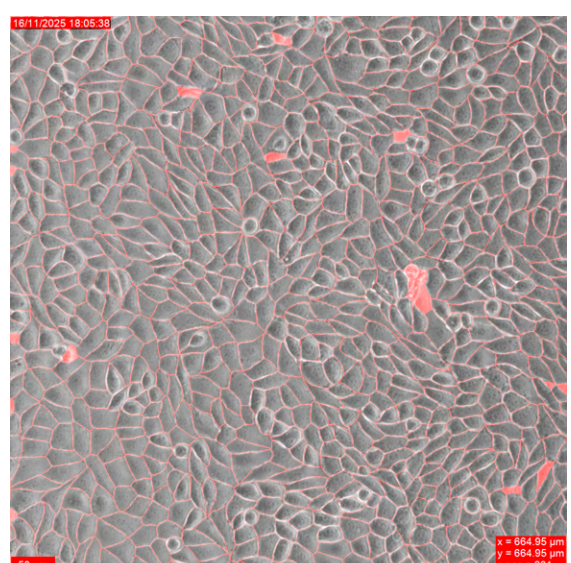
b)



c)



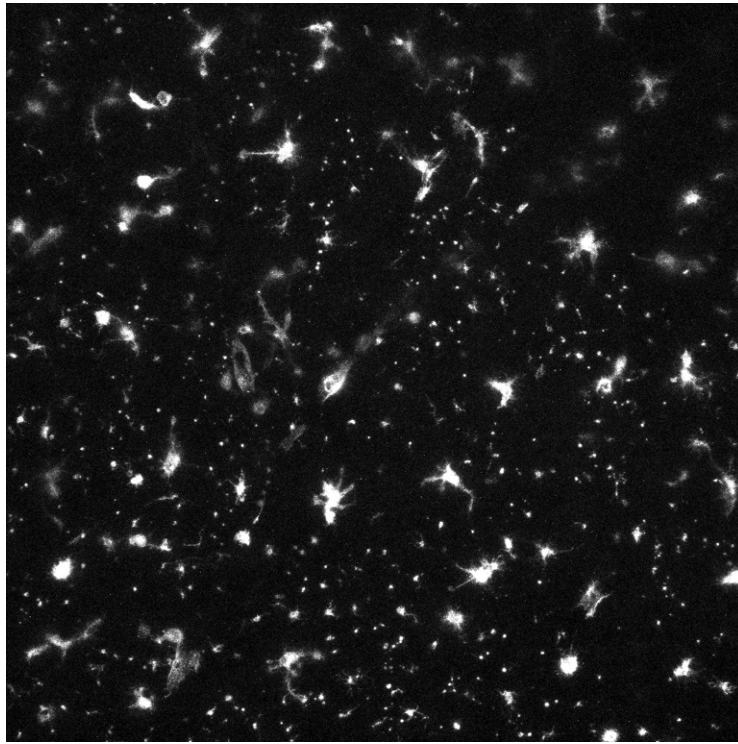
d)



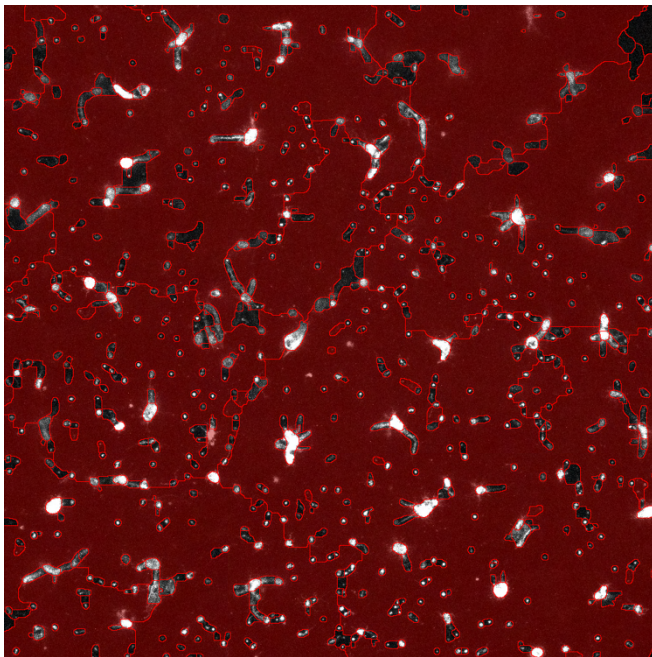
e)

Figure 7. F6 results. a) Original image, b) Intensity Thresholding, c) ImageJ Sobel + Watershed, d) Mask R-CNN, e) Developed Sobel+Watershed, f) Cellpose-SAM

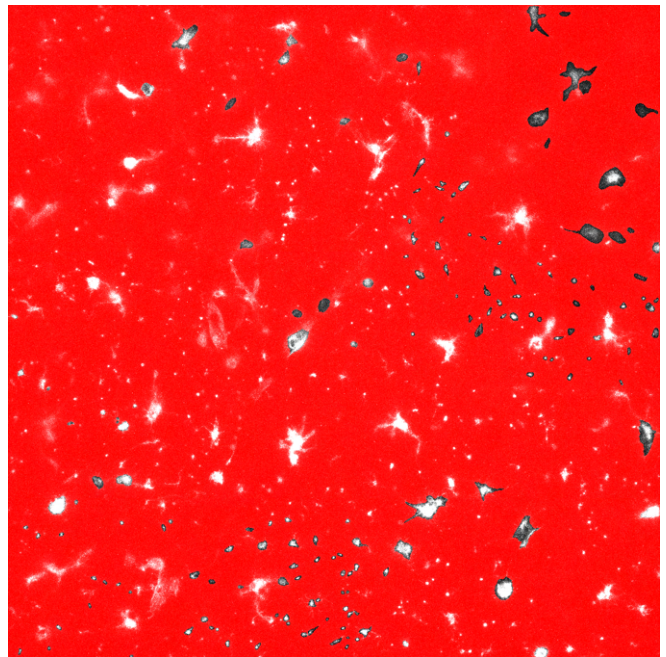
8. APPENFIX 8 – M1 SEGMENTATION RESULTS



a)



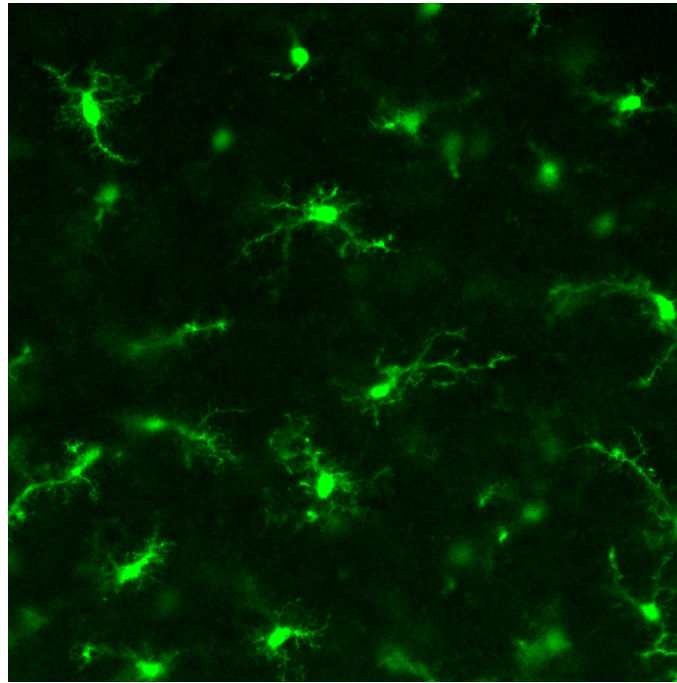
b)



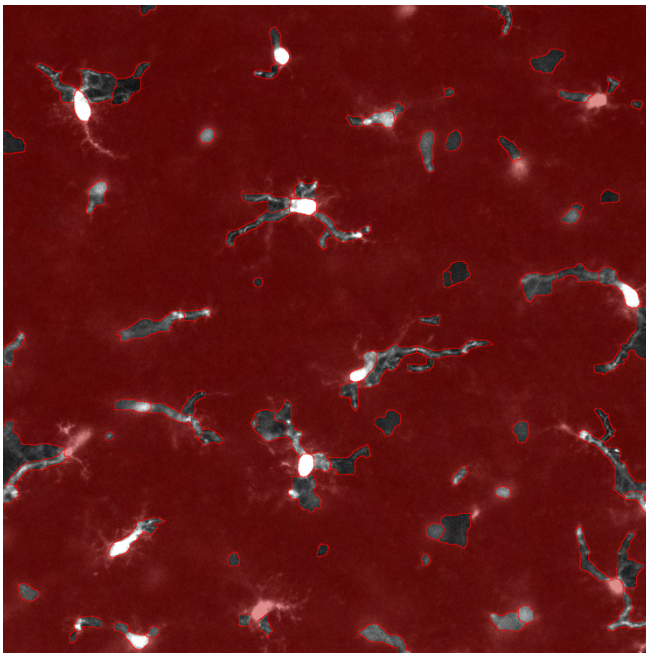
c)

Figure 8. M1 results. a) Original image, b) Developed Sobel+Watershed, c) Cellpose-SAM

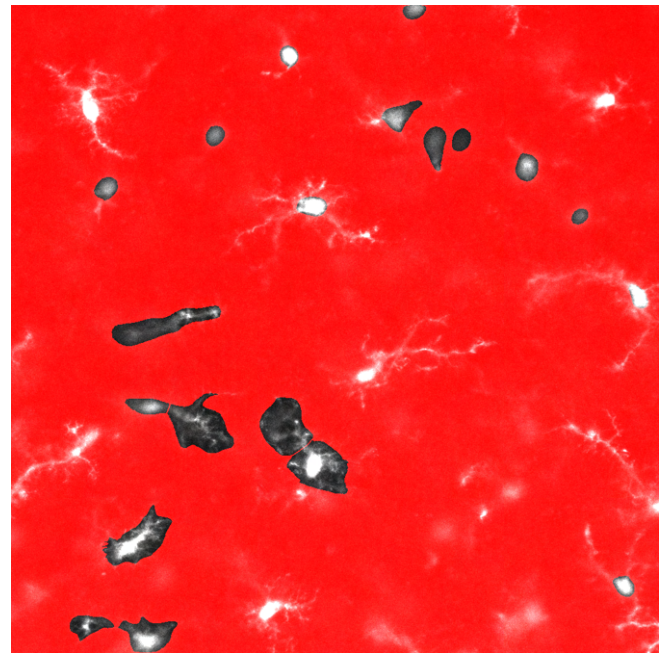
9. APPENDIX 9 – M2 SEGMENTATION RESULTS



a)



b)



c)

Figure 9. M2 results. a) Original image, b) Developed Sobel+Watershed, c) Cellpose-SAM

10. APPENDIX 10 – INCORPORATING DEVELOPED SOBEL + WATERSHED INTO GUI

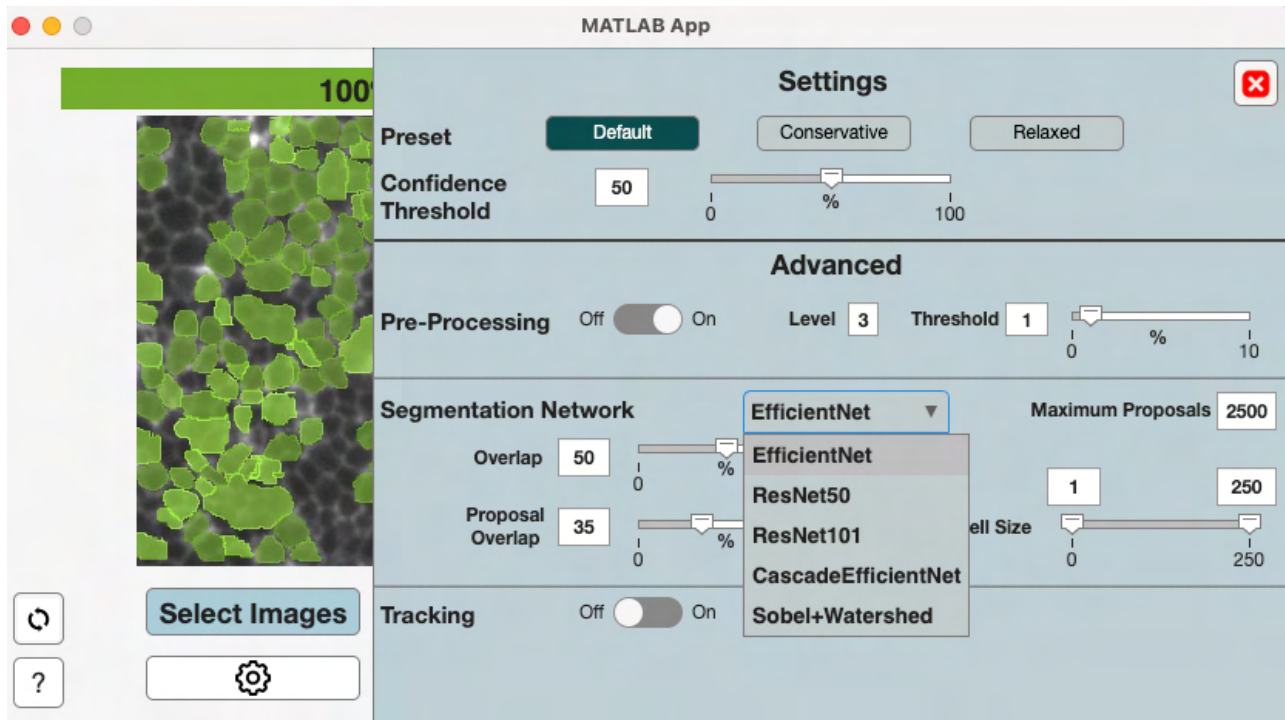


Figure 10. S+W algorithm added into the drop down menu among other Deep Learning algorithms.

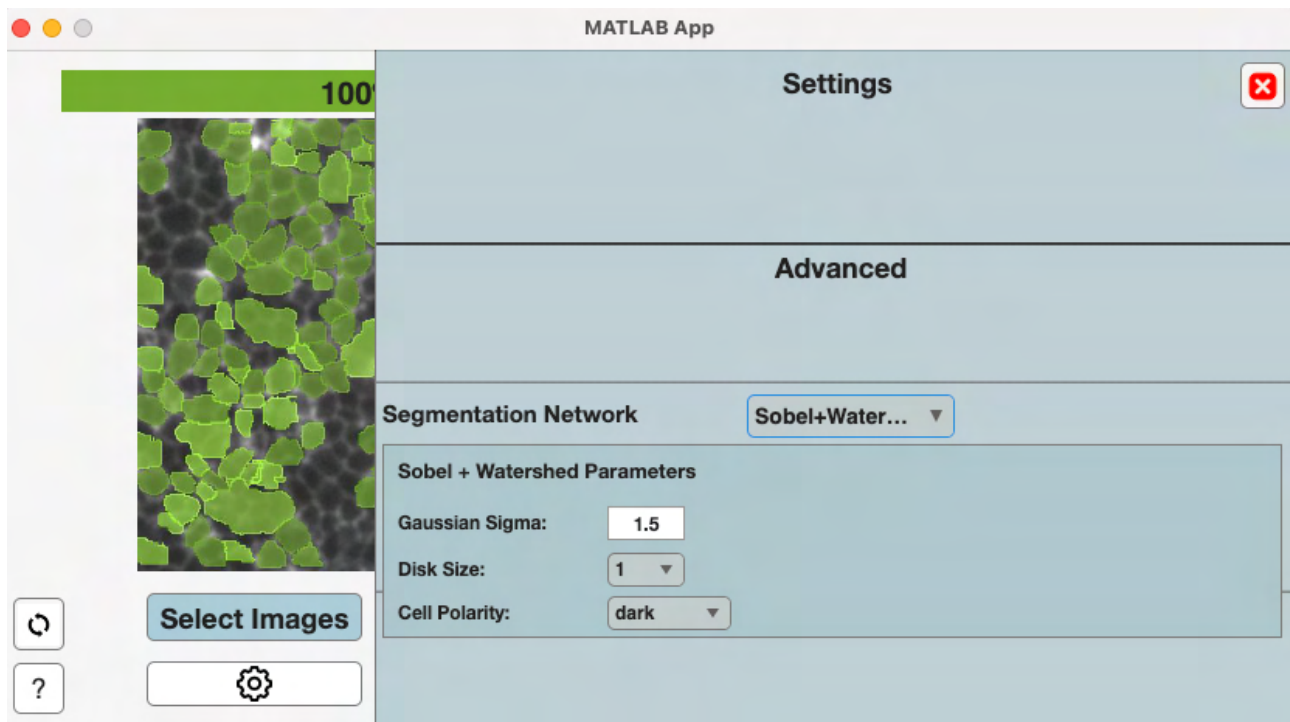


Figure 11. User input panel developed for when Sobel + Watershed algorithm is chosen.

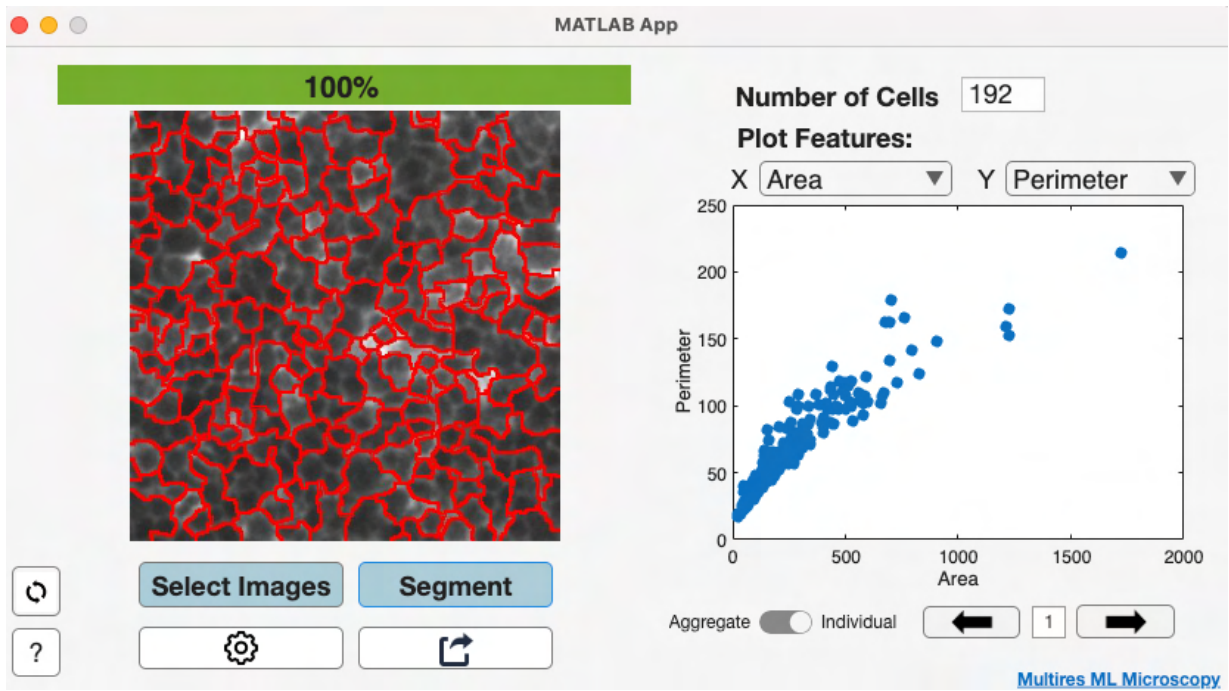


Figure 12. S+W segmentation completed in the GUI.